Build the Recommender

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Build the Recommender

> Prepare Build the Recommender: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Implement a **deterministic recommender**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/24, ai-2/33. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Project Problem and Success Test

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

Look for: Define a small application purpose

Reference example, not the only acceptable wording: Suggest a suitable eligible activity for a teacher-supervised session; the teacher reviews and decides.

Agents and Tools

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher **explanation**: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Predictive vs Generative

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Reference example, not the only acceptable wording: Teacher **explanation**: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering.

Show why other books are excluded.

Concept to explain

A transparent **baseline** makes later improvements understandable.

Term	Meaning
baseline	A simple reference method used to judge whether a more complex method adds value.
candidate	An item or proposed answer being considered before a final selection.
weighted total	A sum formed after multiplying each contribution by its assigned weight.
explanation	An account connecting a result to the mechanism, calculation or evidence that supports it.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; **explanation** 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 24 as a two-session investigation using fictional data. Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No. Filter by the hard constraint before ranking.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Apply the connected idea: Building the data pipeline

Teach Week 32 as a two-session investigation using fictional data. Build the validation **stage** of the planned chooser first.

A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5. The batch **stage** must separately check name uniqueness.

Only accepted records may reach the Week 24 selection function. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

Validate, filter eligibility, calculate scores, resolve ties, then report with limits.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can implement **deterministic** filtering, scoring and ties for a complete catalogue case.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Build the Recommender using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each **stage**, ask what changed and how you could test it.

Which **stage** could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Weights: genre=3, difficulty=2, length=1.

2. Compute every **candidate**'s terms and total.

3. Filter hard constraints, sort remaining totals high to low.

4. **Output** top choice, score breakdown, one limitation, and a tie rule.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

The teaching case

Compute the winner and **evidence** to display.

Exact explanation to model

> **Why the result follows**
>
> A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

Explain the added worked example

Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields.

Explain the added worked example · 1

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run.

After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

Key terms

Term	Meaning
Recommender	A procedure that suggests options using a stated policy
Tie	Two options have the same comparison score
Deterministic	The same inputs and rules produce the same result
Explanation	Evidence describing how a result was obtained

Apply the connected idea: Building the data pipeline

Apply the connected idea: Building the data pipeline · 2

Build the validation **stage** of the planned chooser first. A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5.

The batch **stage** must separately check name uniqueness. Only accepted records may reach the Week 24 selection function.

Apply the connected idea: Building the data pipeline · 3

The worked calls print True then False. Short-circuit checks keep absent or wrongly typed fields away from later operations.

This Boolean helper does not explain the reason; add reason codes in the reporting **stage** or use a validation result record. Do not describe a range failure as a no-match outcome.

The first passes. Replacing minutes with 61 fails the range. Exit: No. The batch uniqueness contract fails. Block the ambiguous batch and report duplicate names for correction.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Predict the result if the list order is reversed, and if the limit is 8.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal **explanation** together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Build the Recommender

A fictional book chooser filters a request before comparing preference scores.

The facts we will use

Supplied facts

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

Supplied facts

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

AI ACADEMY / AI-2/33

Filtering before scoring

	Pages	Preference score	Status
A	90	2	eligible
B	70	1	eligible
C	60	3	subject fails
D	80	3	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/33

Build the Recommender

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Filter

A,B,D meet subject and page requirements; C fails subject.

2. Compute scores

A=2; B=1; D=3.

3. Choose best

Return D with both preference matches recorded.

4. Deterministic output

Reordering the input rows must not change D's unique winning score.

CHANGE ONE THING

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Filter	A,B,D meet subject and page requirements; C fails subject.
2. Compute scores	A=2; B=1; D=3.
3. Choose best	Return D with both preference matches recorded.
4. Deterministic output	Reordering the input rows must not change D's unique winning score.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

What this example does not establish

Keep the stated limit

Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

Change one thing in the pictured case

Change one thing

Change: Change D’s pages to 91, leaving every other field unchanged.

Predicted result

Prediction: D becomes ineligible, so A wins with score 2.

Why the result follows

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Build the Recommender, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

2. State the decision to explain

Write the question to explain: Compute the winner and **evidence** to display.

3. Reveal the worked reasoning

Reveal after **prediction**s: A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

Hold back the independent task

Keep for an independent attempt: A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an **explanation**.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Reversing still returns A. Limit 8 leaves no eligible item and returns None. Exit: X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: State recommend([], 10) and the worked items with limit 8.

Reasoning to listen for

Teacher check: Both return None; the caller should display no eligible option.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.
>
> Display the chosen rule and let a person review whether an option is suitable.

Collect **explanation**s, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group **explanation** while others solve a more demanding variation.

Connected teaching cycle: Building the data pipeline

Target: Separate validation selection and reporting

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

> Completion practice: What does valid_item return if learning is absent?

Reasoning to listen for

Teacher check: False: get returns None and the exact integer check fails.

Individual check: For Building the data pipeline, use its worked case to evaluate this proposed **explanation**: “A successful run proves the conclusion”. Explain the deciding step.

Expected repair

Expected correction: Each **stage** should have a clear **input** and **output** contract. A program can be correct at one **stage** but wrong at their connection, such as passing rejected rows into selection.

Keep acceptance and rejection counts in the report.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Explain why B does not win after D is excluded even though it has fewer pages.

Reasoned teacher answer

Page count is an eligibility limit, not a ranking preference here. A scores 2 and B scores 1.

Changed-case reasoning: D becomes ineligible, so A wins with score 2. The strict excess over the 90-page maximum changes filtering before ranking.

Collect each learner’s **explanation** before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Build the Recommender investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Build the editable score table, hand-check one row, run two profiles, and compare system ranking with independent recalculation.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an **explanation**. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

A convenient shortcut preserves meaning: Updating best on score >= best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties.

Test both orders and a no-match case after the repair.

Check these misconceptions

A successful run proves the conclusion: Each **stage** should have a clear **input** and **output** contract. A program can be correct at one **stage** but wrong at their connection, such as passing rejected rows into selection.

Keep acceptance and rejection counts in the report.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an **explanation**. Fresh retry: X (minutes 9, fun 2, learning 3), Y (10,3,2), Z (11,6,1), limit 10.

Accept X/Y, reject Z rating; select Y score 8 versus X 7.

A convenient shortcut preserves meaning: Calling the selector on raw data can crash or produce a result from invalid ratings. Route data through validation first.

Duplicate names require a batch check even when each row individually passes; block the ambiguous batch for review instead of arbitrarily keeping one.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral **explanation**, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Build the Recommender questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

> **Teacher key · Worked case answer**

A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

Independent question

Independent question · withhold the key

A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

Expected independent answer

Teacher key · Expected independent answer

Neither matches both conditions. Return no exact match and ask whether to relax a preference; do not present A or B as a full match.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The system must recommend something even when constraints fail.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Only A meets genre and length constraints; its score is 3. B exceeds the page limit and C fails genre. Do not claim A beat eligible competitors when none existed. Include no-match and missing-field handling in the prototype.

Explanations for the additional workbook examples

W1 Trace selection

Workbook question · W1

Why does A win the worked example even though B is **process**ed first?

Teacher reasoning · W1

Both score 8 and qualify; A is alphabetically first under the tie rule.

W2 Handle no match

Workbook question · W2

State recommend([], 10) and the worked items with limit 8.

Teacher reasoning · W2

Both return None; the caller should display no eligible option.

Teacher reasoning · W2

For the empty list, the loop has no iterations, so best stays None. With the worked durations 10 and 9 under a limit of 8, neither item passes the eligibility check, so best again stays None.

The caller must turn this result into a clear no-eligible-option message; it must not treat None as an activity name.

W3 Repair order dependence

Workbook question · W3

A tied selection changes after reversing **input** order. Identify the likely missing policy and test.

Teacher reasoning · W3

An explicit **deterministic** tie rule is missing or incorrect. Apply alphabetical name order and rerun both orders.

W4 Implement independently

Workbook question · W4

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

Teacher reasoning · W4

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None.

W5 Explain the result

Workbook question · W5

Write a one-sentence **explanation** of X winning without calling it objectively best.

Teacher reasoning · W5

X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

Building the data pipeline — W1 Trace a validation

Explain why a row with fun "4" fails while fun 4 can pass.

The first is text; the schema requires an exact integer rating from 0 to 5.

Building the data pipeline — W2 Check missing fields

What does valid_item return if learning is absent?

False: get returns None and the exact integer check fails.

When learning is absent, row.get("learning") returns None. The condition type(value) is not int is true, so the helper returns False before doing the range comparison.

This is a schema rejection; it is different from a valid item that later exceeds the selection time limit.

Building the data pipeline — W3 Repair the connection

A program validates rows but still sends the original raw list to recommend. What should

> change?

Pass only the accepted, uniqueness-checked list. Preserve rejection information separately for reporting.

Building the data pipeline — W4 Build independently

Combine validation and the Week 24 selector for A (minutes 10, fun 3, learning 2), B (12,4,1), and C (8,"5",5), limit 10. State accepted rows, rejected row, selection and **explanation**.

A and B validate; C fails fun type. B exceeds the limit. A wins with score 8. Report 2 accepted of 3, one rejection and one eligible item. Verify unique names.

Building the data pipeline — W5 Handle duplicates

> Two valid rows have the same name but different ratings. Should individual validation alone permit selection?

No. The batch uniqueness contract fails. Block the ambiguous batch and report duplicate names for correction.

Assess independent transfer

> **Independent question · withhold the key**
>
> Selected case: Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated.
>
> Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

Selected case answer

> **Teacher key · Selected case answer**
>
> D fails length and F fails genre. E scores3, G and H score5. H wins the tie with 90 versus95 pages. Filtering is separate from scoring; this written preference rule is not proof of a trained model.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Build the Recommender to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

This is the complete **baseline**: **input**s, scoring, filter, ranking, **explanation**, and abstain rule.

Keep the decision, **input**, result, **explanation** and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 34: Test Diverse Users. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Trace best and best_score row by row. Extension: return a record with name, score and reason while retaining the same selection policy.

Independent W4 **evidence**: X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None. Record whether the learner needed no help, a hint or **explanation**.

Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

Next week: File **input** and **output**. Reuse this week's explicit contracts and **evidence**.

Readiness and reassessment for Building the data pipeline

Connect paper **stage** cards with **input/output** labels. Extension: return specific reasons for missing, type and range errors without exposing personal data.

Independent W4 **evidence**: A and B validate; C fails fun type. B exceeds the limit. A wins with score 8. Report 2 accepted of 3, one rejection and one eligible item. Verify unique names.

Record whether the learner needed no help, a hint or **explanation**. Fresh retry: X (minutes 9, fun 2, learning 3), Y (10,3,2), Z (11,6,1), limit 10. Accept X/Y, reject Z rating; select Y score 8 versus X 7.

Plan the delayed return

Suggested return points in this level: ai-2/34, ai-2/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Build the Recommender lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Implement a **deterministic recommender**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: State recommend([], 10) and the worked items with limit 8.

Supported practice answer

Reasoned practice answer: Both return None; the caller should display no eligible option.

Check every learner before discussion: Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken **explanation** or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change D’s pages to 91, leaving every other field unchanged. Require a **prediction** and a reason before comparing. Expected result: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner’s **explanation** and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an **explanation**. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-**explanation** links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

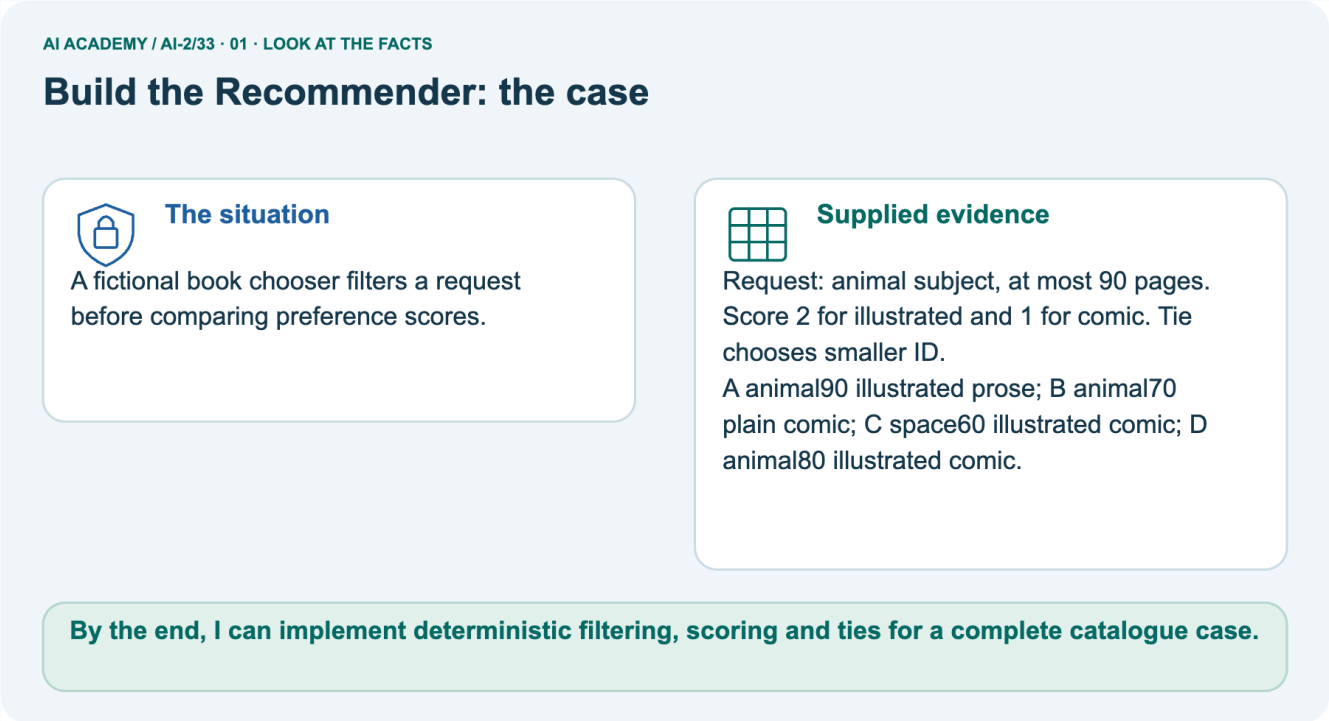
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess **explanation**, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Build the Recommender graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Build the Recommender: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book chooser filters a request before comparing preference scores. Supplied facts: Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

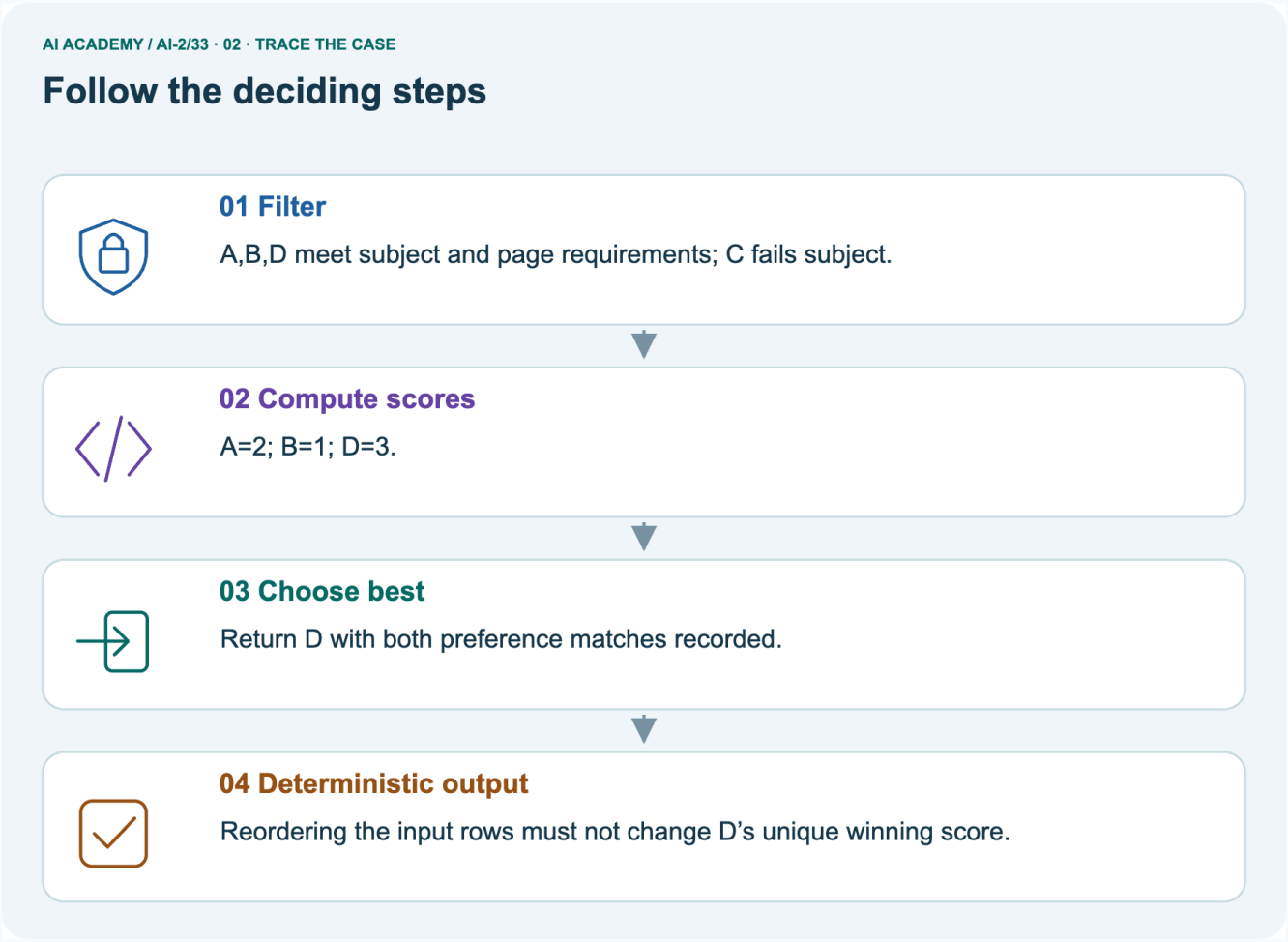
A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

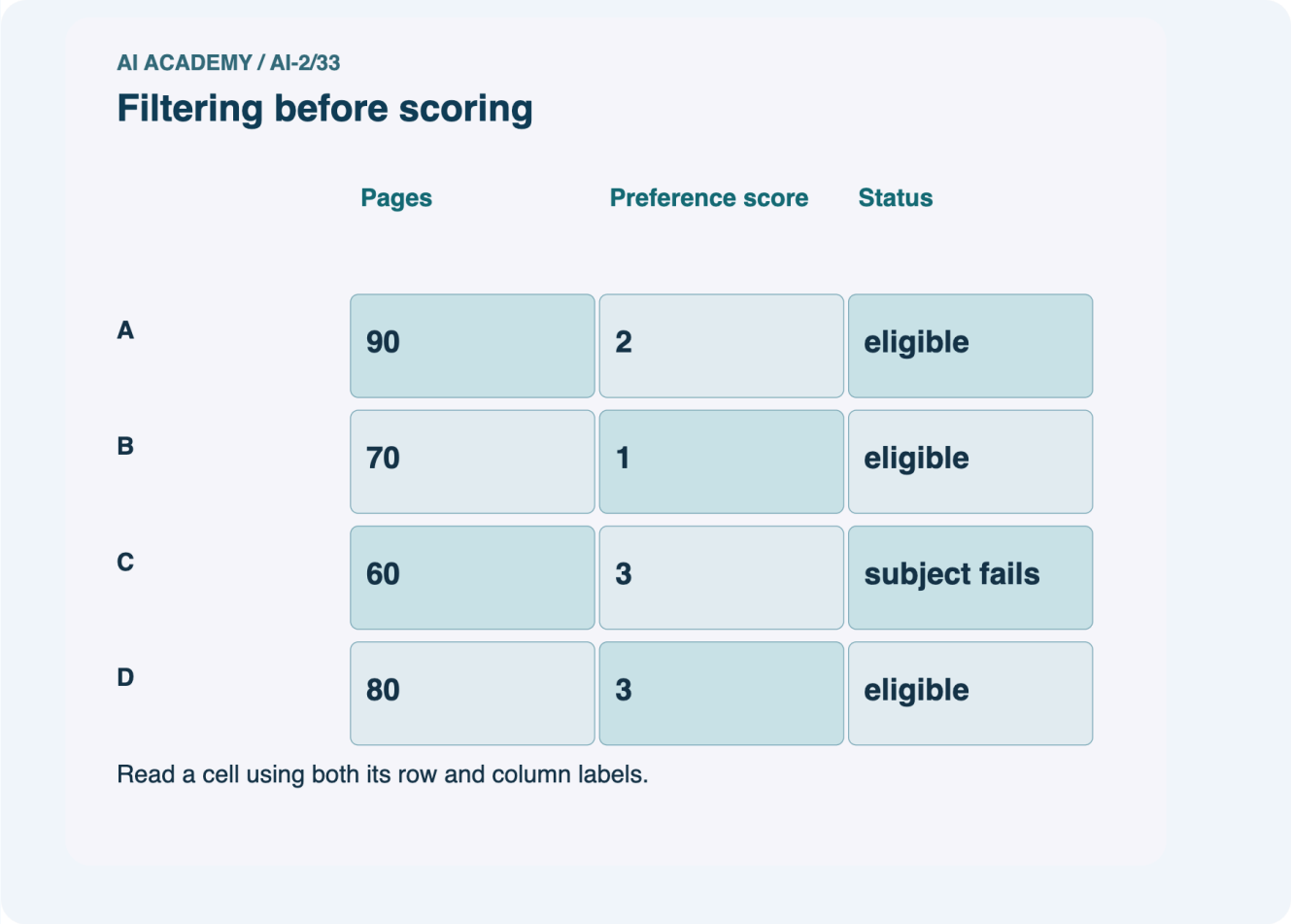
Filter: A,B,D meet subject and page requirements; C fails subject. → Compute scores: A=2; B=1; D=3. → Choose best: Return D with both preference matches recorded. → **Deterministic output**: Reordering the **input** rows must not change D's unique winning score.

> Point to each numbered **stage** and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered **stage** and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Filtering before scoring



Follow the labelled picture with the matching explanation. Open the vector diagram

Filtering before scoring. A: Pages = 90, Preference score = 2, Status = eligible. B: Pages = 70, Preference score = 1, Status = eligible. C: Pages = 60, Preference score = 3, Status = subject fails.

D: Pages = 80, Preference score = 3, Status = eligible.

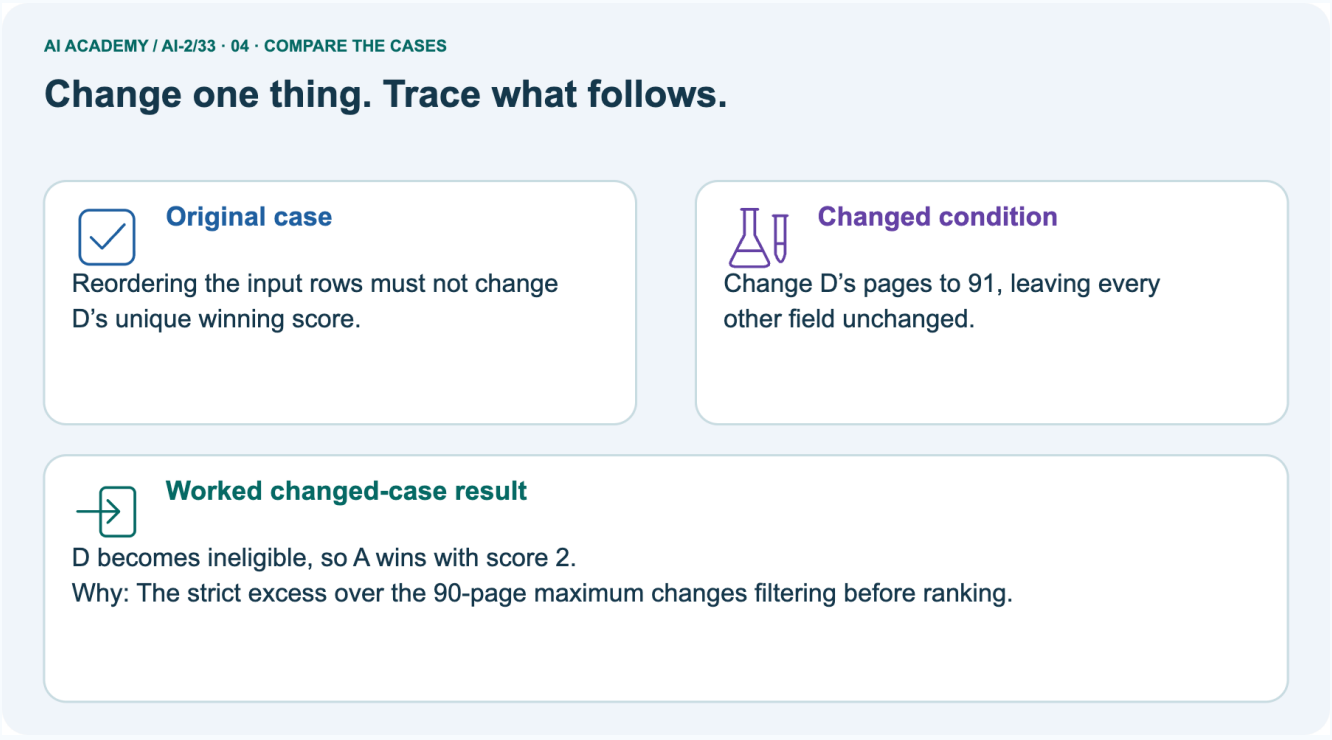
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Reordering the **input** rows must not change D's unique winning score. Changed condition: Change D's pages to 91, leaving every other field unchanged. Expected result: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

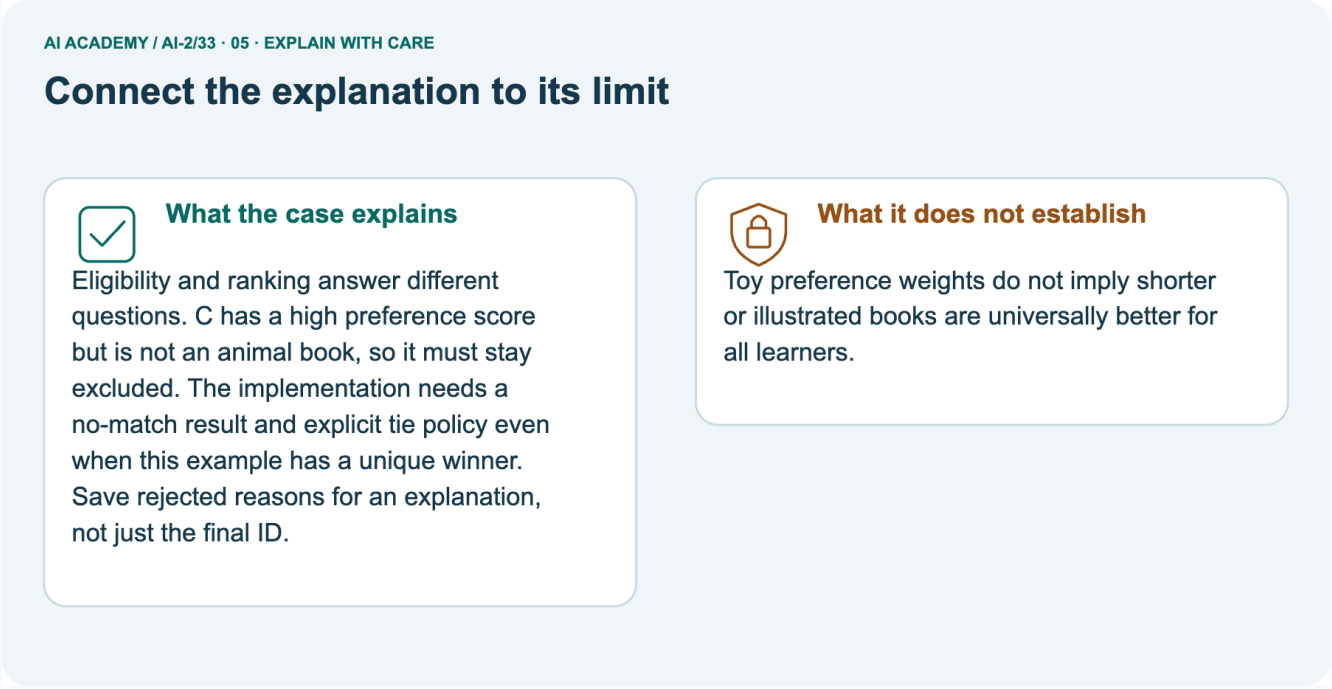
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The strict excess over the 90-page maximum changes filtering before ranking.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

Limit: Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

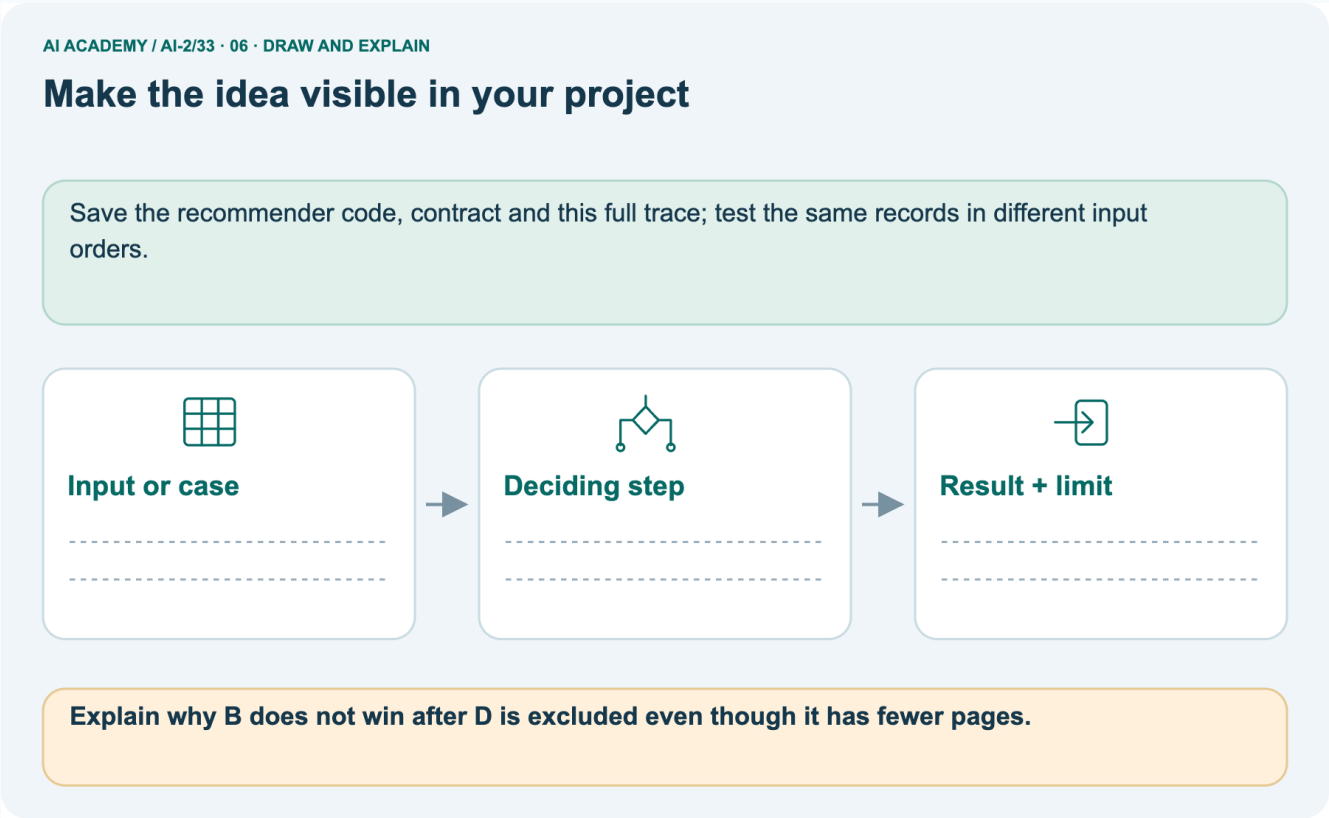
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B does not win after D is excluded even though it has fewer pages.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Page count is an eligibility limit, not a ranking preference here. A scores 2 and B scores 1.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.